Computer Coding

S.T.E.E.L

S.T.E.E.L Program

Astro Clare Technology

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Welcome to the S.T.E.E.L Program

I am excited to meet you and watch you take the journey into becoming an engineer. In this program, you will learn skills such as coding, problem-solving, teamwork, communication, debugging, and many more. At the end of the first quarter of this program, you will have your very own website! How exciting is that?

If you have no knowledge of coding or programming, **DON'T WORRY**. I have designed each website and section of coding to be beginner-friendly to tailor the needs of everyone involved. So, whether you have a computer, laptop, phone, or tablet, you'll be able to build your own website portfolio.

Additionally, you'll have access to long-term support and resources, ensuring that you continue to grow and refine your skills even after the program ends. You'll not only gain technical skills but also learn how to present yourself professionally in the tech industry.

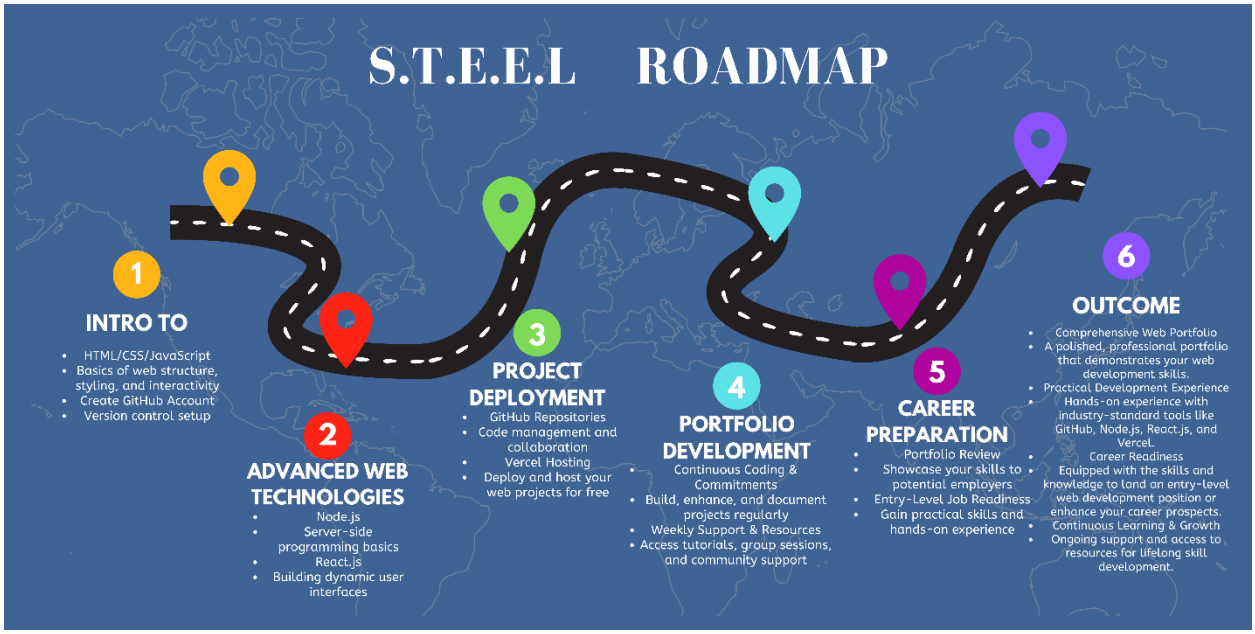
Roadmap Overview

Here's a quick look at what we'll cover throughout the program:

1. **Intro to Web Basics**
    - HTML/CSS/JavaScript: Basics of web structure, styling, and interactivity.
    - Create GitHub Account: Version control setup.
2. **Advanced Web Technologies**
    - Node.js: Server-side programming basics.
    - React.js: Building dynamic user interfaces.
3. **Project Deployment**
    - GitHub Repositories: Code management and collaboration.
    - Vercel Hosting: Deploy and host your web projects for free.
4. **Portfolio Development**
    - Continuous Coding & Commitments: Build, enhance, and document projects regularly.
    - Weekly Support & Resources: Access tutorials, group sessions, and community support.
5. **Career Preparation**
    - Portfolio Review: Showcase your skills to potential employers.
    - Entry-Level Job Readiness: Gain practical skills and hands-on experience.

6. **Outcome**

- Comprehensive Web Portfolio: A polished, professional portfolio that demonstrates your web development skills.
- Practical Development Experience: Hands-on experience with industry-standard tools like GitHub, Node.js, React.js, and Vercel.
- Career Readiness: Equipped with the skills and knowledge to land an entry-level web development position or enhance your career prospects.
- Continuous Learning & Growth: Ongoing support and access to resources for lifelong skill development.



Intro To Web Basics

HTML

In our HTML section we will cover the fundamentals of Web Development, creating our websites skeleton. We will use elements, tags, in-line styling and properties to create our skeleton.

Example: In this example we create a navigation menu to access different areas of our website.

```
<header>
  <nav>
    <ul>
      <li><a href="#home">Home</a></li>
      <li><a href="#about">About Me</a></li>
      <li><a href="#portfolio">Portfolio</a></li>
      <li><a href="#contact">Contact</a></li>
    </ul>
  </nav>
</header>
```

CSS

In our Cascading Style Sheet (or CSS file) we will cover the fundamentals of creating a UI/UX (User Interface and User Experience) that is eye catching and modern. This will give you a creative edge in your future web portfolio against your peers.

Example: In this example we use our navigation menu to target links to style by giving them different properties per selector.

```
header nav ul li a {
  color: #fff;
  text-decoration: none;
  font-weight: bold;
  transition: color 0.3s ease;
}
```

JavaScript

In our JavaScript section we will cover some basic programming fundamentals of using the JavaScript scripting programming language to add some interactivity to our website. This will not be

the end of JavaScript for this project though. We will later cover how to migrate a feature to toggle light and dark mode into our site to be user friendly for those with conditions to sensitive colors or lighting.

Example: In this example we are creating a smooth scroll function for the “click” event when navigation is being used.

```
document.querySelectorAll('a[href^="#"]').forEach(anchor => {
  anchor.addEventListener('click', function(e) {
    e.preventDefault();


    document.querySelector(this.getAttribute('href')).scrollIntoView({
      behavior: 'smooth'
    });
  });
});
```

Version Control Accounts

In this section we will create version control accounts (GitHub and Vercel). These will be used to allow us to host our new portfolios for free online, accessible to all. Other Coders and Programmers will even be able to see your code which will be awesome when collaborating with other Software Engineers!

Advanced Web Technologies

Node and React JS

In this section we will discover and discuss what Node.JS and React.JS are. In shorthand:

Node.JS is a runtime environment used to run JavaScript on a server, outside of our browser. It’s mainly used to build scalable applications such as web servers.

React.JS is a JavaScript Library used for building user interfaces, mainly single page applications. This allows developers to create reusable User Interface components and manage the states of the application efficiently.

Project Deployment

In this section we will learn how to set up our GitHub Repositories and manage our code along with collaborating with our peers. From here we will be able to see every change in our code, update and current release of our project.

After we have covered GitHub, we will then attach our Vercel account to our GitHub account allowing us to use our repository as the backbone for our new website that will be hosted on Vercel.

Career Development

We will go through a career development process where we will do a portfolio review, practice job readiness, and learn how to market ourselves as newly found engineers.

1. Portfolio Review

- **Description:** Offer personalized portfolio reviews as a core feature of the program. Emphasize that you will receive feedback on your portfolio to ensure you effectively showcase your skills to potential employers.
- **Implementation:**
    - Include scheduled review sessions where instructors or industry professionals provide detailed feedback.
    - Host "Portfolio Showcase" events where you will present your portfolios to peers and invited guests, simulating real-world job application processes.

2. Entry-Level Job Readiness

- **Description:** Focus on equipping students with the practical skills and experience needed for entry-level positions in tech and engineering fields.
- **Implementation:**
    - Integrate project-based learning where students work on real-world problems or simulations.
    - Include workshops on resume building, interview preparation, and job search strategies.
    - Offer mock interviews or role-playing scenarios to practice professional interactions.

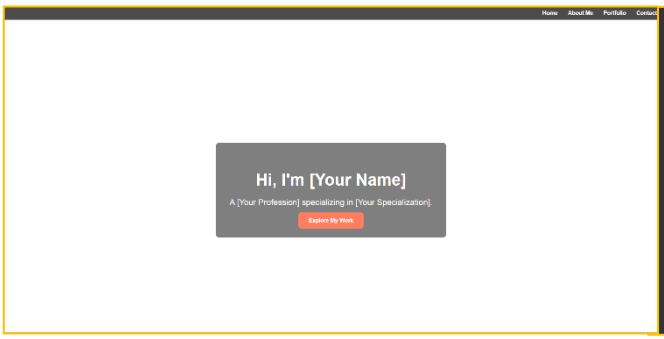
3. Marketing These Elements

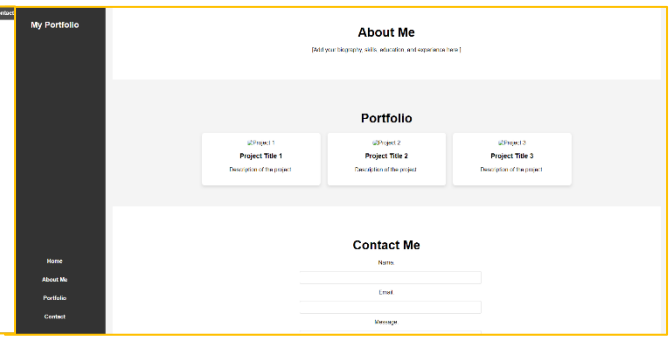
- **Promotional Content:** Highlight these features in your marketing materials, such as your website, brochures, and social media.
- **Testimonials and Success Stories:** Gather and showcase testimonials from individuals who asked you to build them a website, web app or webpage (optional).

Outcome

Upon completing the S.T.E.E.L Program, you will achieve the following:

- **Comprehensive Web Portfolio:** A polished, professional portfolio that demonstrates your web development skills.
- **Practical Development Experience:** Hands-on experience with industry-standard tools like GitHub, Node.js, React.js, and Vercel.
- **Career Readiness:** Equipped with the skills and knowledge to land an entry-level web development position or enhance your career prospects.
- **Continuous Learning & Growth:** Ongoing support and access to resources for lifelong skill development, including advanced resources, open-source contributions, hackathons, and coding communities.





Ready to Take the Next Step?

Join the S.T.E.E.L. program today and start building your future in web development. Whether you're a beginner or looking to enhance your skills, we provide the tools, support, and guidance you need to succeed.

Contact Us

- **Website:** https://s-t-e-e-l.vercel.app
- **Email:** scottclarence5@gmail.com
- **Phone:** (810) 241-8724
- **Social Media:** Follow us on Facebook Group | YouTube

Get Started Now

Visit our website to learn more and enroll. Your future in web development can begin with S.T.E.E.L.